

EXPERIMENT 38

Malware Activity Correlation

AIM

To correlate process, file, and network events that occur close together and identify suspicious activity sequences.

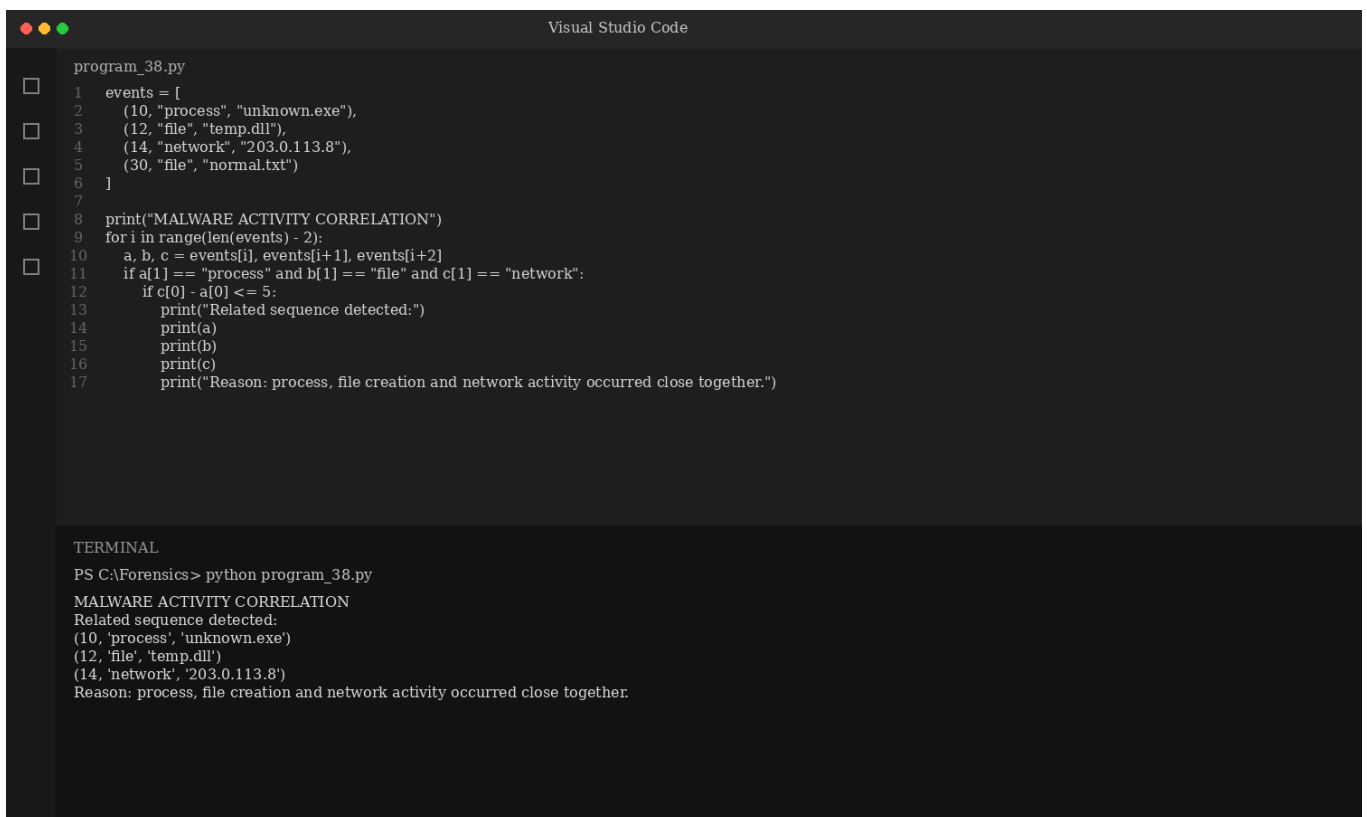
ALGORITHM

1. Create simulated process, file, and network events.
2. Convert timestamps into minutes.
3. Compare events occurring within a small time window.
4. Link suspicious process, file creation, and outbound connection.
5. Produce a correlation report.

CODE

```
events = [  
    (10, "process", "unknown.exe"),  
    (12, "file", "temp.dll"),  
    (14, "network", "203.0.113.8"),  
    (30, "file", "normal.txt")  
]  
  
print("MALWARE ACTIVITY CORRELATION")  
for i in range(len(events) - 2):  
    a, b, c = events[i], events[i+1], events[i+2]  
    if a[1] == "process" and b[1] == "file" and c[1] == "network":  
        if c[0] - a[0] <= 5:  
            print("Related sequence detected:")  
            print(a)  
            print(b)  
            print(c)  
            print("Reason: process, file creation and network activity occurred close together.")
```

OUTPUT



The screenshot shows the Visual Studio Code editor with a file named `program_38.py`. The code defines a list of events and a loop that checks for suspicious sequences. The terminal output shows the execution of the script, which prints the events and identifies a related sequence of process, file creation, and network activity.

```
program_38.py  
1  events = [  
2      (10, "process", "unknown.exe"),  
3      (12, "file", "temp.dll"),  
4      (14, "network", "203.0.113.8"),  
5      (30, "file", "normal.txt")  
6  ]  
7  
8  print("MALWARE ACTIVITY CORRELATION")  
9  for i in range(len(events) - 2):  
10     a, b, c = events[i], events[i+1], events[i+2]  
11     if a[1] == "process" and b[1] == "file" and c[1] == "network":  
12         if c[0] - a[0] <= 5:  
13             print("Related sequence detected:")  
14             print(a)  
15             print(b)  
16             print(c)  
17             print("Reason: process, file creation and network activity occurred close together.")
```

TERMINAL

```
PS C:\Forensics> python program_38.py  
MALWARE ACTIVITY CORRELATION  
Related sequence detected:  
(10, 'process', 'unknown.exe')  
(12, 'file', 'temp.dll')  
(14, 'network', '203.0.113.8')  
Reason: process, file creation and network activity occurred close together.
```

RESULT

The program correlates nearby process, file and network events to identify a suspicious activity sequence.